

Performance & Evaluation of Classifiers

Almost everything about

confusion matrix



What is a confusion matrix?

It is a tool to evaluate performance of supervised machine learning algorithm when used for classification problem.

Consider....

You train a ML model for a data set that is labelled only for YES or NO for each data row.

Ex. Loan worthiness of a person against their income, education, credit score, marital status, number of children and other features.

But remember....

Only for this problem our classes are YES and NO. In general classes can be any two different things such as Positive or Negative, Bus or Truck, Spam or Not Spam, Dog or Cat and so on.

NOTE: Classes do not have to be necessarily opposite to each other.

*Now how do you check the performance of your model on **test data**?*

*With the help of **confusion matrix**.*

x1	x2	Y (actual)	Y (predicted)
-	-	0	1
-	-	1	1
-	-	0	0
-	-	1	1
-	-	1	1
-	-	0	1
-	-	1	0

Let us assume your *test data* contained *100* rows. Then your confusion matrix might look something like this...

		<i>Yes Predicted</i>		No
<i>Actual</i>	Yes	45	15	
	No	10	30	

$n = 100$

*Ok, so what
does these
numbers mean?*

Oh before that let us consider

YES to be positive label

No to be negative label

		Yes Predicted	No
Actual	Yes	45	15
	No	10	30

TP ✓

$n = 100$

True Positives

When model predicts YES
and actual value is also YES

		Yes Predicted	No
Actual	Yes	45	15
	No	10	30

$n = 100$

FP !

False Positives or Type 1 error

When model predicts YES
but actual value is NO

	Yes	Predicted	No
Yes	45		15
Actual			
No	10		30

$n = 100$

TN ✓

True Negatives

When model predicts **NO**
and actual value is also **NO**

		Yes Predicted No	
Actual	Yes	45	15
	No	10	30

$n = 100$

FN !

False Negatives or Type II error

When model predicts NO
but actual value is YES

In summary...

		Yes	<i>Predicted</i>	No
<i>Actual</i>	Yes	TP		FN
	No	FP		TN

n = total no. of predictions

No more confusion...

What was the prediction?

True Positive

True Negative

False Positive

False Negative



Was the prediction right?

Ok!

***But how does this help
in evaluating model
performance?***

Classification models can be evaluated using ...

- Accuracy
- Precision
- Area Under Curve (AUC)
- Recall
- F1 Score

Accuracy

Total number of correct/true predictions among all the predictions.

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{n}$$

Note: Used for balance dataset

Precision

Measures classifiers' exactness.

Positive cases correctly predicted
among all predicted positive cases.

$$\text{precision} = \frac{TP}{TP + FP}$$

Note:

- Denotes that out of all predicted result how many of them are correctly predicted
- Used for imbalance dataset, ex; mail (spam or not spam)

Recall

Measures classifiers' completeness.

Positive cases correctly predicted
among all actual positive cases.

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Note:

- Denotes that out of all the actual results how many are correctly predicted results
- Used for imbalance dataset, ex; healthcare (disease or not disease)

F-beta Score

A machine learning metric that combines precision and recall into a single score by using a beta parameter, denoted as β .

$$\text{precision} = (1 + \beta^2) \frac{\text{precision} \times \text{recall}}{\beta^2 * \text{precision} + \text{recall}}$$

Note:

- Used when both FP and FN are important.
- Used for imbalance dataset, ex; stock market prediction

$$\text{precision} = (1 + 1^2) \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

Both FN & FP are important, $\beta = 1$

$$\text{precision} = (1 + (0.5)^2) \frac{\text{precision} \times \text{recall}}{0.5 * \text{precision} + \text{recall}}$$

FN >> FP, $\beta = 0.5$

$$\text{precision} = (1 + 2^2) \frac{\text{precision} \times \text{recall}}{2 * \text{precision} + \text{recall}}$$

FN << FP, $\beta = 2$

F1 Score

Judging the model on accuracy, precision or recall alone is flawed since most of the ML or data science problems are context based.

Therefore, F1 Score is usually a safe metric to evaluate a model. Model with highest F1 Score not only returns true positive but also increases the number of true positives.

$$\text{F1 score} = \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \times 2$$

AUC

Before coming to area under curve, it is important we cover the following.....

- Sensitivity
- Specificity
- ROC curve
(Sensitivity vs 1-Specificity)

Sensitivity

It is the True Positive Rate or simply recall.

Best value = 1

Worst value = 0

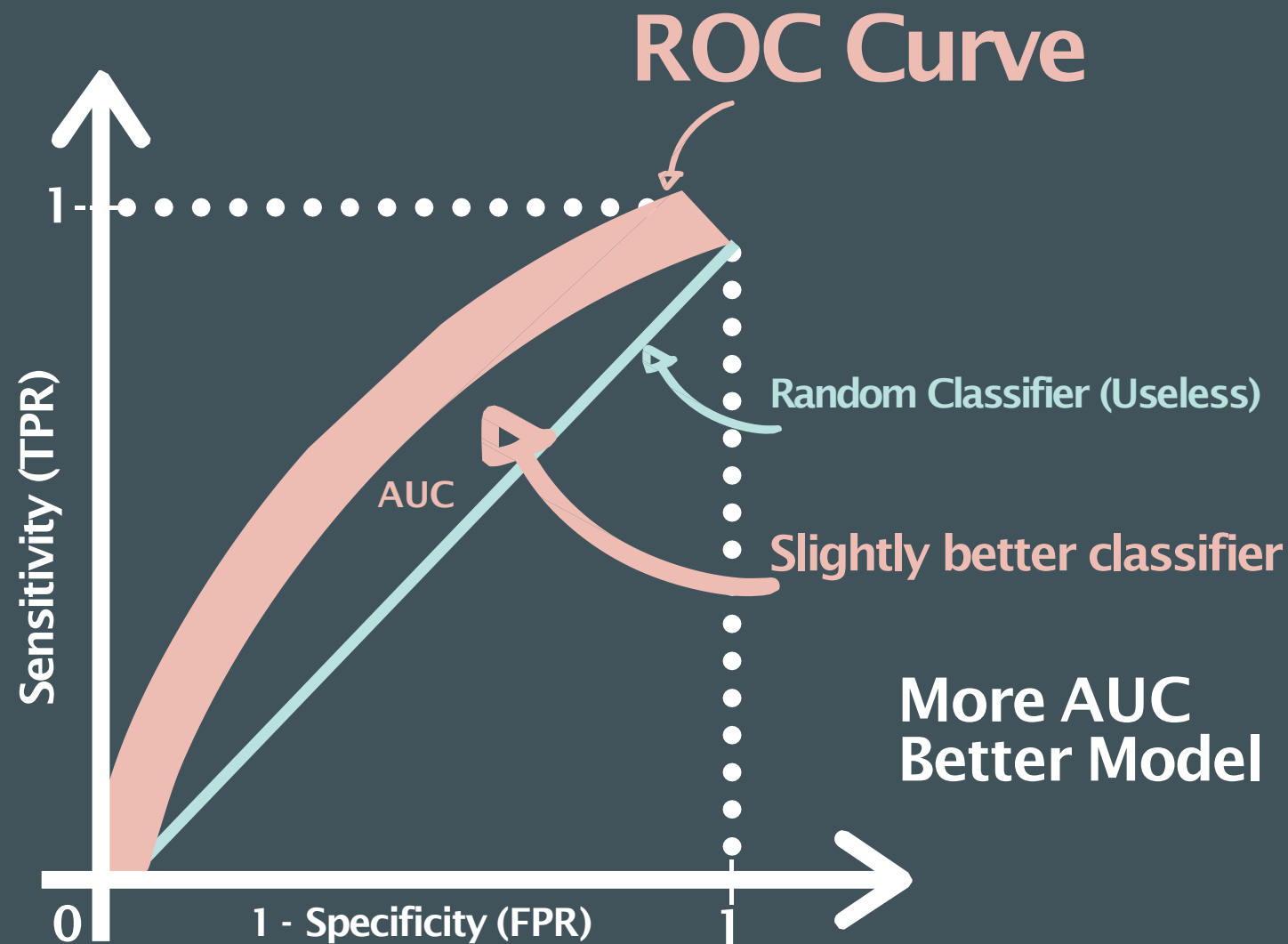
Specificity

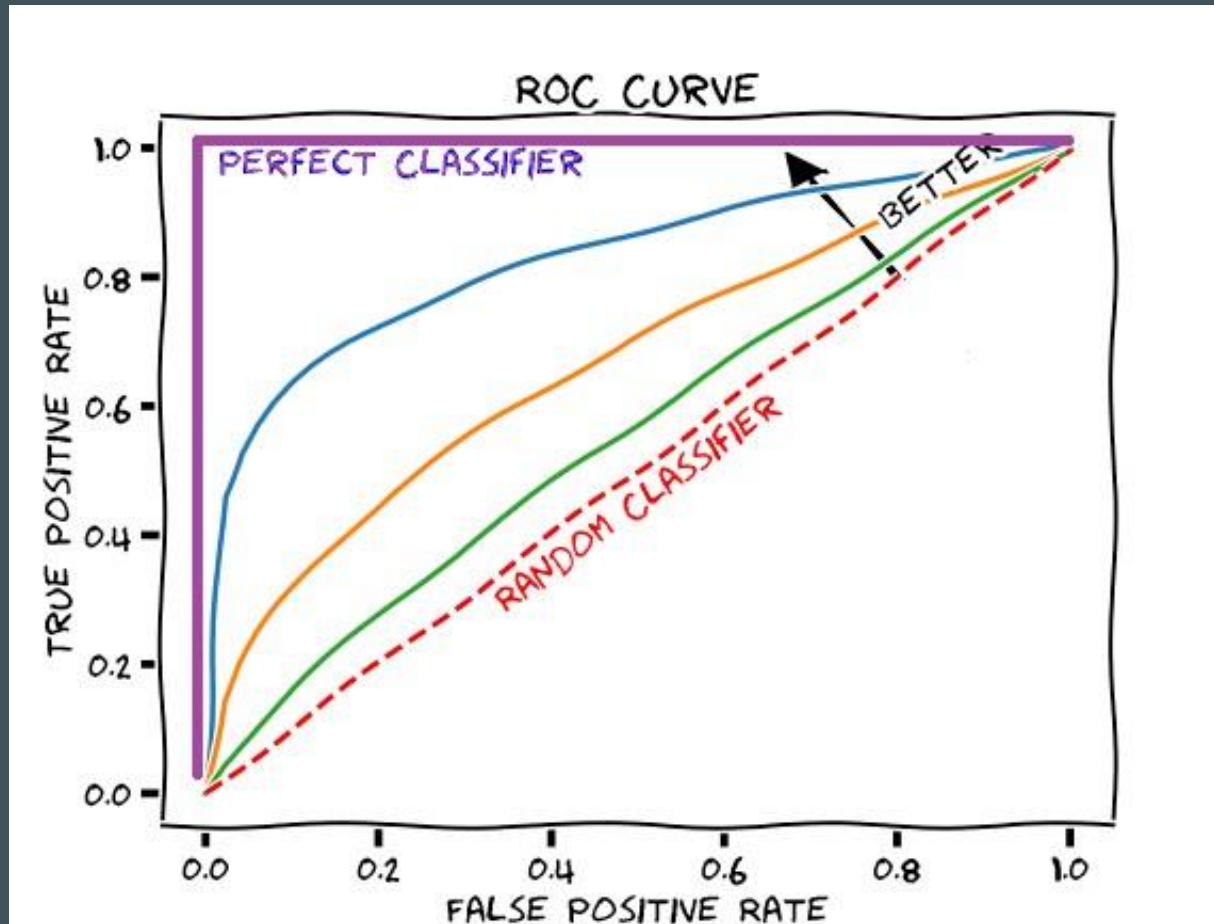
Number of correct negative predictions divided by the total number of negatives
 $= \text{TN} / (\text{TN} + \text{FP})$

It is also called true negative rate (TNR).

False Positive Rate = $1 - \text{Specificity}$

1	0.8
0	0.96
1	0.4
1	0.3
0	0.2
1	0.7





Pic source: glassboxmedicine.com

Points to remember...

- Confusion matrix is a classifier evaluation tool.
- Classes need not be opposites.
- One class is labelled positive, other is labelled negative. Just labelled, does not mean anything.
- Accuracy, recall and precision depends on the context and may not be true judge of the model.
- F1 Score and AUC are more comprehensive evaluation metrics.

Thank You!